Ali **R. Vahdati**

Evolutionary biologist · Computational biologist

Zurich, Switzerland

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Key Skills

Machine Learning — Applied advanced machine learning and deep learning techniques to understand large datasets, make predictions, and analyze high-throughput molecular data. Demonstrated proficiency in both supervised and unsupervised machine learning methods.

Scientific Modeling — **Proficient in creating and interpreting complex biological models, with a sound understanding of biology and immunology. Applied these skills to understand and predict biological phenomena, contributing to advancements in fields such as human evolution, population genetics, and ecology.**

Statistics — Utilized advanced statistical methods to interpret and analyze data from various biological research projects, facilitating informed decision-making.

Mathematical Optimization — **Employed mathematical optimization techniques, such as linear programming, in biological modeling and data analysis, optimizing solutions for complex biological processes.**

Experience with Large Datasets — Experienced in handling and analyzing large genomic datasets from high-throughput technologies, facilitating the extraction of meaningful insights and advancing research in evolutionary biology.

Population Genetics — **Applied knowledge of population genetics in various research projects, contributing to the understanding of genetic diversity and evolution.**

Programming — Proficient in multiple programming languages and tools including R, Python, Julia, Nim, LaTeX, MySQL, SQLite, Git, Continuous integration, Unit testing, and Docker. Experience in creating reproducible research and statistical programming. Familiar with Linux command line and high-performance computing environments, enabling the development of efficient algorithms and models for biological research.

Data Visualization — **Skilled in using tools like Vega-lite, Matplotlib, and Makie.jl, as well as visualization principles to create effective visual representations of complex data, aiding in data interpretation and communication of research findings.**

Soft Skills — Demonstrated resilience, leadership, commitment, and adaptability in various research and teaching roles, contributing to successful project outcomes and effective team collaborations.

Transferable Skills — **Proven ability in teaching, presentation, and writing, with experience in effectively communicating complex data and research findings to diverse audiences including biologists, clinicians, and data scientists. Proficient in time management and project management, enhancing the effectiveness of research processes and facilitating knowledge transfer in academic settings.**

Languages — English (fluent), Persian (native), German (B1)

Work Experience

Senior lecturer
Zurich, Switzerland

Prof. Zollikofer group, Department of Anthropology, University of Zurich — *Aug. 2020 - Current*

- Conducted extensive research on ancient human dispersals and extinctions using advanced simulation techniques. My work has provided valuable insights into human evolutionary history.

Consultant *Busan, South Korea*

IBS center for climate physics, University of Pusan *May and June 2022*

- Developed a comprehensive model to analyze the impact of climate on human genetic diversity. This model is being used for multiple projects.

Postdoctoral researcher *Zurich, Switzerland*

Prof. Zollikofer group, Department of Anthropology, University of Zurich *Sep. 2017 - Aug. 2020*

- Led multiple research projects, utilizing statistical tools and machine learning techniques to analyze and interpret data. My work resulted in several significant findings, contributing to the broader field of anthropology and computational biology.

Modeler *Madeira, Portugal*

University of Madeira *2022-2023*

- Constructed a model for understanding metacommunity dynamics of complex life-cycles in exploited seascapes.

Modeler *Mérida, Mexico*

Centro de investigación Cientifica de Yucatán *Oct. 2021*

- Developed statistical models to understand interaction strengths in food webs.

Education

University of Zurich *Zurich, Switzerland*

PhD in Evolutionary Biology *Jan. 2012 - Sep. 2017*

- Advisor: Prof. Dr. Andreas Wagner, Institute of Evolutionary Biology and Environmental Studies.
- Thesis: Genotype networks: convergent evolution, population size and adaptation.
- Published four scholarly papers.

University of Manchester *Manchester, UK*

MSc in Evolutionary Biology and Genomics *Sep. 2010 - Sep. 2011*

- Advisor 1: Prof. Dr. Reinmar Hager, Faculty of Life Sciences.
- Thesis 1: Identifying QTL and genomic imprinting effects on mice tissue phenotypes.
- Advisor 2: Prof. Dr. Casey Bergman, Faculty of Life Sciences.
- Thesis 2: Analysis of replication timing on genetic variation in the human genome.
- Published one scholarly paper.

Selected Peer-reviewed Papers

EvoDynamics.jl: a framework for modeling eco-evolutionary dynamics *Journal of Open Source Software*

R Vahdati A, Melián C J *2022*

Exploring Late Pleistocene hominin dispersals, coexistence and extinction with agent-based multi-factor models *Quaternary Science Reviews*

R Vahdati A, Weissmann JD, Timmermann A, Ponce de León MS, Zollikofer CPE *2022*

Agents.jl: A performant and feature-full agent based modelling software of minimal code complexity *Simulation*

Datseris G, R Vahdati A, DuBois T C *2022*

Drivers of Late Pleistocene human survival and dispersal: an agent-based modeling and machine learning approach *Quaternary Science Reviews*

R Vahdati A, Weissmann JD, Timmermann A, Ponce de León MS, Zollikofer CPE *2019, 221*

Agents.jl: agent-based modeling framework in Julia *Journal of Open Source Software*

R Vahdati, A *2019, 4(42), 1611*

Population size affects adaptation in complex ways: simulations on empirical adaptive landscapes. *Evolutionary Biology*

Vahdati AR, Wagner A *2017*

Effect of Population Size and Mutation Rate on the Evolution of RNA Sequences on an Adaptive Landscape Determined by RNA Folding — *International Journal of Biological Sciences*
Vahdati A R, Sprouffske K, Wagner A — *2017;13(9):1138–51*

Parallel or convergent evolution in human population genomic data revealed by genotype networks — *BMC Evolutionary Biology*
Vahdati A R, Wagner A — *2016*

VCF2Networks: applying Genotype Networks to Single Nucleotide Variants data — *Bioinformatics*
Dall'Olio G M, Vahdati A R, Bertranpetit J, Wagner A, Laayouni H — *2014*

Genomic imprinting and genetic effects on muscle traits in mice — *BMC Genomics*
Kaerst S*, Vahdati A R*, Brockmann G, Hager R — *2012*
* Equal contribution

Teaching Experience

BIO397 - Applied Machine Learning - block course — *University of Zurich*
Lecturer — *2019-2022*
- I have designed and delivered lectures and exercises on Julia programming language, linear algebra, multivariate calculus, machine learning algorithms, neural networks, and evaluation and improvement of machine learning models.
- I have prepared students with no previous knowledge of machine learning to be able to design and perform complete machine learning workflows.

An introduction to agent-based modeling — *University of Pusan, Busan, Switzerland*
Lecturer — *2022*
- I gave a lecture and hands-on tutorial on using Julia language for building agent-based models.

Reproducible research - workshop — *EAWAG – Kastanienbaum, Switzerland*
Lecturer — *2018*
- I gave a lecture and hands-on tutorial on using Git version control software and collaborating using Github.